Assignment 4: UML Use Case Modeling

a. Identify/discover a use case
   - Solve Sudoku Puzzle

b. Brief description for the use case

   This use case allows a User to pass a text file containing a partially completed 9×9 Sudoku grid to the system. The system then parses the file, prints the initial grid, exhaustively calculates all valid solutions using depth-first search with backtracking, and displays the solutions.

c. Outline of the events for the use case
   1. Start application.
   2. Provide puzzle filename.
   3. Validate and load puzzle file.
   4. Display original puzzle.
   5. Calculate puzzle solutions.
   6. Display results.
   7. Terminate application.

d. Detailed description

   1. Start application: The User executes the Sudoku Solver program in the command line environment.

   2. Provide puzzle filename: The User provides the target text file.

   3. Validate and load puzzle file: The system attempts to open the specified text file. It reads the file character by character, translating underscores (_) into empty cells (0) and digit characters (1-9) into their respective integer values within an internal 9×9 grid array.

   4. Display original puzzle: The system outputs the name of the file to the console, followed by a printout of the initial, unsolved 9×9 grid for user verification.

   5. Calculate puzzle solutions: The system initiates a recursive depth-first search with backtracking. It identifies an empty cell and tests candidate digits, ensuring no candidate violates Sudoku row, column, or 3×3 block constraints. Valid candidates are placed, and the system recurses. If a path fails, it backtracks and tries the next candidate.

6. Display results: Each time the system successfully fills all 81 cells, it increments a solution counter and prints the fully solved grid. If the entire search space is exhausted and the solution counter remains at zero, the system prints "No solution found."

7. Terminate application: After all solutions are displayed, the application closes file streams and exits.

UML Use Case Diagram:

